

Comprehensive Notes on DC Circuit Theorems: Maximum Power Transfer Theorem

1. High-Level Overview & Core Intuition

What is Maximum Power Transfer Theorem?

The **Maximum Power Transfer Theorem** states that a linear two-terminal circuit delivers the maximum possible amount of power to an adjustable (variable) load resistance R_L if and only if the load resistance is set exactly equal to the equivalent internal resistance of the network (the Thevenin resistance, R_{Th}).

$$\text{Condition for Maximum Power Transfer: } R_L = R_{Th}$$

Physical Intuition: Why $R_L = R_{Th}$?

Consider power delivered to the load resistor: $P_L = I^2 \cdot R_L$.

- If R_L is extremely small ($R_L \rightarrow 0$):**
 - Current I is high, but the resistance R_L receiving the power is almost zero.
 - Power consumed $P_L = I^2 \cdot (0) = 0$ W.
- If R_L is extremely large ($R_L \rightarrow \infty$):**
 - Resistance is huge, but the current drops to zero ($I \rightarrow 0$).
 - Power consumed $P_L = (0)^2 \cdot R_L = 0$ W.
- The Sweet Spot ($R_L = R_{Th}$):**
 - Maximum power transfer occurs at the optimal balance between high current and high resistance.

Mathematical Proof & Derivation

From Thevenin's theorem, the current in the loop is:

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

The power P_L delivered to the load resistor R_L is:

$$P_L = I_L^2 \cdot R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 \cdot R_L = \frac{V_{Th}^2 \cdot R_L}{(R_{Th} + R_L)^2}$$

To find the maximum power, set the derivative of P_L with respect to R_L equal to zero ($\frac{dP_L}{dR_L} = 0$):

$$\frac{dP_L}{dR_L} = V_{Th}^2 \cdot \left[\frac{(R_{Th} + R_L)^2 - R_L \cdot 2(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] = 0$$

Simplifying the numerator:

$$(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0$$

$$(R_{Th} + R_L) \cdot [(R_{Th} + R_L) - 2R_L] = 0$$

$$R_{Th} - R_L = 0 \implies R_L = R_{Th}$$

Formula for Maximum Power (P_{max})

Substituting $R_L = R_{Th}$ into the power equation:

$$P_{max} = \frac{V_{Th}^2 \cdot R_{Th}}{(R_{Th} + R_{Th})^2} = \frac{V_{Th}^2 \cdot R_{Th}}{(2R_{Th})^2} = \frac{V_{Th}^2 \cdot R_{Th}}{4R_{Th}^2}$$
$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Efficiency at Maximum Power Transfer

At maximum power transfer ($R_L = R_{Th}$), total power supplied by the source is:

$$P_{source} = I_L^2 \cdot (R_{Th} + R_L) = I_L^2 \cdot (2R_{Th})$$

Efficiency (η) is the ratio of output power to total source power:

$$\eta = \frac{P_{load}}{P_{source}} = \frac{I_L^2 \cdot R_{Th}}{I_L^2 \cdot (2R_{Th})} = \frac{1}{2} = 50\%$$

- Key Takeaway:** At the maximum power transfer condition, the circuit efficiency is always 50%. Half the power is dissipated inside the source network (R_{Th}), and half is transferred to the load (R_L).

Circuit Schema Representation

 Diagram: Maximum Power Transfer Network

- Visual Description:**
 - Left Side:** A block labeled "Two terminal linear circuit" with output terminals A and B connected across a variable resistor labeled R_L (indicated by an arrow through the resistor).
 - Right Side:** Equivalent Thevenin circuit showing a DC voltage source V_{Th} connected in series with internal resistance R_{Th} across terminals A and B . The load resistor attached across A and B is explicitly set to $R_L = R_{Th}$ to achieve P_{max} .

2. Step-by-Step Problem-Solving Methodology

- Identify the Variable Load Resistor (R_L or R):** Disconnect R_L from terminals A and B .
- Find R_{Th} :** Deactivate all independent sources (short-circuit voltage sources, open-circuit current sources) and compute the equivalent resistance looking into terminals A and B .
- Set $R_L = R_{Th}$:** This gives the exact load resistance value required for maximum power absorption.
- Find V_{Th} :** Calculate the open-circuit voltage V_{AB} across terminals A and B using standard circuit analysis techniques.
- Calculate P_{max} :** Evaluate using $P_{max} = \frac{V_{Th}^2}{4R_{Th}}$.

3. Comprehensive Worked Examples

Illustration 1: Bridge Circuit Network

Problem Statement:

The variable resistor ' R ' in the circuit shown is adjusted until it absorbs the maximum power from the circuit. Calculate the value of ' R ' for maximum power and determine the maximum power absorbed by ' R '.

- **Visual Description:**

- Diamond bridge network with a central vertical 40 V battery.
 - Top-left resistor: 80 Ω .
 - Top-right resistor: 20 Ω .
 - Bottom-left resistor: 10 Ω .
 - Bottom-right resistor: 90 Ω .
 - The variable resistor R Ω is connected across the top vertex and bottom vertex of the diamond.
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Step-by-Step Solution

Step 1: Disconnect Variable Resistor R

Disconnect R to define upper terminal A (top node) and lower terminal B (bottom node).

Step 2: Find Thevenin Resistance (R_{Th})

1. Short-circuit the 40 V source in the central branch.
2. Redraw network relative to terminals A and B :
 - Top-left 80 Ω and bottom-left 10 Ω resistors become connected in parallel:

$$R_{left} = 80 \parallel 10 = \frac{80 \times 10}{80 + 10} = \frac{800}{90} = 8.89 \Omega$$

- Top-right 20 Ω and bottom-right 90 Ω resistors become connected in parallel:

$$R_{right} = 20 \parallel 90 = \frac{20 \times 90}{20 + 90} = \frac{1800}{110} = 16.36 \Omega$$

3. Total input resistance R_{Th} across A and B :

$$R_{Th} = R_{left} + R_{right} = \frac{800}{90} + \frac{1800}{110} = 8.889 + 16.364 = 25.25 \Omega$$

- **Value of R for Maximum Power:**

$$R = R_{Th} = 25.25 \Omega$$

Step 3: Find Open-Circuit Voltage (V_{Th})

1. Set the bottom node B or middle node as reference.
2. The 40 V source fixes the voltage across the left and right legs.
3. Left branch potential divider (top to bottom):

$$V_{A1} = 40 \text{ V} \times \frac{80}{80 + 10} = 40 \times \frac{80}{90} = 35.56 \text{ V}$$

4. Right branch potential divider:

$$V_{A2} = 40 \text{ V} \times \frac{20}{20 + 90} = 40 \times \frac{20}{110} = 7.27 \text{ V}$$

5. Open-circuit voltage across A and B :

$$V_{Th} = V_A - V_B = 35.56 \text{ V} - 7.27 \text{ V} = 28.29 \text{ V}$$

Step 4: Calculate Maximum Power (P_{max})

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(28.29)^2}{4 \times 25.25} = \frac{800.32}{101} \approx 7.92 \text{ W}$$

Illustration 2: Complex Multi-Loop Circuit with Star Network

Problem Statement:

Determine the value of the load resistance to be connected across terminals A & B such that maximum power is delivered to the load. Also, determine the maximum power.

Diagram: Multi-loop Network with Star Resistors

- **Visual Description:**
 - Far-left branch: A 2 A independent current source directed downwards.
 - Parallel $6\text{ }\Omega$ resistor.
 - Middle section: A star (Y) connected resistor configuration with three $2\text{ }\Omega$ resistors.
 - Bottom branch connecting star node to left branch: $6\text{ }\Omega$ resistor.
 - Middle-right branch: $6\text{ }\Omega$ resistor in series with an 8 V battery.
 - Far-right section: A 6 V battery in series with a $6\text{ }\Omega$ resistor leading to terminal A , and an $8\text{ }\Omega$ resistor connected between terminal A and terminal B .

Step-by-Step Solution

Step 1: Find Thevenin Resistance (R_{Th})

1. Deactivate all independent sources:
 - Open-circuit the 2 A current source.
 - Short-circuit the 6 V and 8 V DC voltage sources.
2. Convert the internal Star (Y) connection of three $2\text{ }\Omega$ resistors into Delta (Δ) to simplify:

$$R_{\Delta} = 2 + 2 + \frac{2 \times 2}{2} = 6\text{ }\Omega \quad \text{for each Delta leg}$$

3. Combining parallel branches across the network systematically yields equivalent resistance looking into A - B :

$$R_{Th} = 4.8\text{ }\Omega$$

- **Value of Load Resistance for Maximum Power:**

$$R_L = R_{Th} = 4.8\text{ }\Omega$$

Step 2: Find Thevenin Voltage (V_{Th})

1. Solve for node voltages across open terminals A and B with active sources (2 A , 6 V , 8 V) using Nodal Analysis.
2. Voltage obtained across open terminals A - B :

$$V_{Th} = 3.6\text{ V}$$

Step 3: Calculate Maximum Power (P_{max})

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(3.6)^2}{4 \times 4.8} = \frac{12.96}{19.2} = 0.675\text{ W} = 675\text{ mW}$$

Illustration 3: Multi-Source Grid Circuit

Problem Statement:

Determine the value of the variable resistor R that will result in maximum power transfer to it from the given circuit. Subsequently, calculate the magnitude of this maximum power.

Diagram: Grid Circuit for Max Power

- **Visual Description:**

- A two-mesh DC network with 100 V battery on right branch and 50 V battery on bottom-left branch.
- Top resistors: 15 Ω and 10 Ω.
- Middle vertical branch resistor: 5 Ω connected to terminal A.
- Center horizontal resistor: 10 Ω.
- Left vertical resistor: 20 Ω.
- Variable resistor R connected vertically between terminal A and terminal B.

Step-by-Step Solution

Step 1: Calculate R_{Th} (Looking into Terminals A and B)

- Short-circuit the 100 V source and 50 V source.
- Remove variable resistor R connected between A and B.
- Reduce resistor combinations:
 - 15 Ω in series with 20 Ω = 35 Ω.
 - $35\ \Omega \parallel 10\ \Omega = \frac{35 \times 10}{35 + 10} = \frac{350}{45} = 7.78\ \Omega$.
 - 7.78 Ω in series with 5 Ω = 12.78 Ω.
 - Parallel combination with right path (10 Ω):

$$R_{Th} = \frac{12.78 \times 10}{12.78 + 10} = \frac{127.8}{22.78} \approx 5.61\ \Omega$$

- **Optimal Resistance Setting:**

$$R = R_{Th} = 5.61\ \Omega$$

Step 2: Calculate V_{Th} Across Open Terminals A and B

- Apply Mesh / Nodal equations with 100 V and 50 V sources active.
- Solving for potential difference $V_{AB} = V_{Th}$:

$$V_{Th} = 68.42\ \text{V}$$

Step 3: Calculate Maximum Absorbed Power

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(68.42)^2}{4 \times 5.61} = \frac{4681.3}{22.44} \approx 208.61\ \text{W}$$

4. Quick Summary & Formula Table

Parameter	Symbol	Governing Formula	Unit
Max Power Condition	R_L	$R_L = R_{Th}$	Ohm (Ω)
Maximum Power Transfer	P_{max}	$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$	Watt (W)
Circuit Efficiency at P_{max}	η	$\eta = 50\%$	Percentage (%)
Source Power at P_{max}	P_{src}	$P_{src} = 2 \cdot P_{max} = \frac{V_{Th}^2}{2R_{Th}}$	Watt (W)